

CSS

(Cascading Style Sheets)

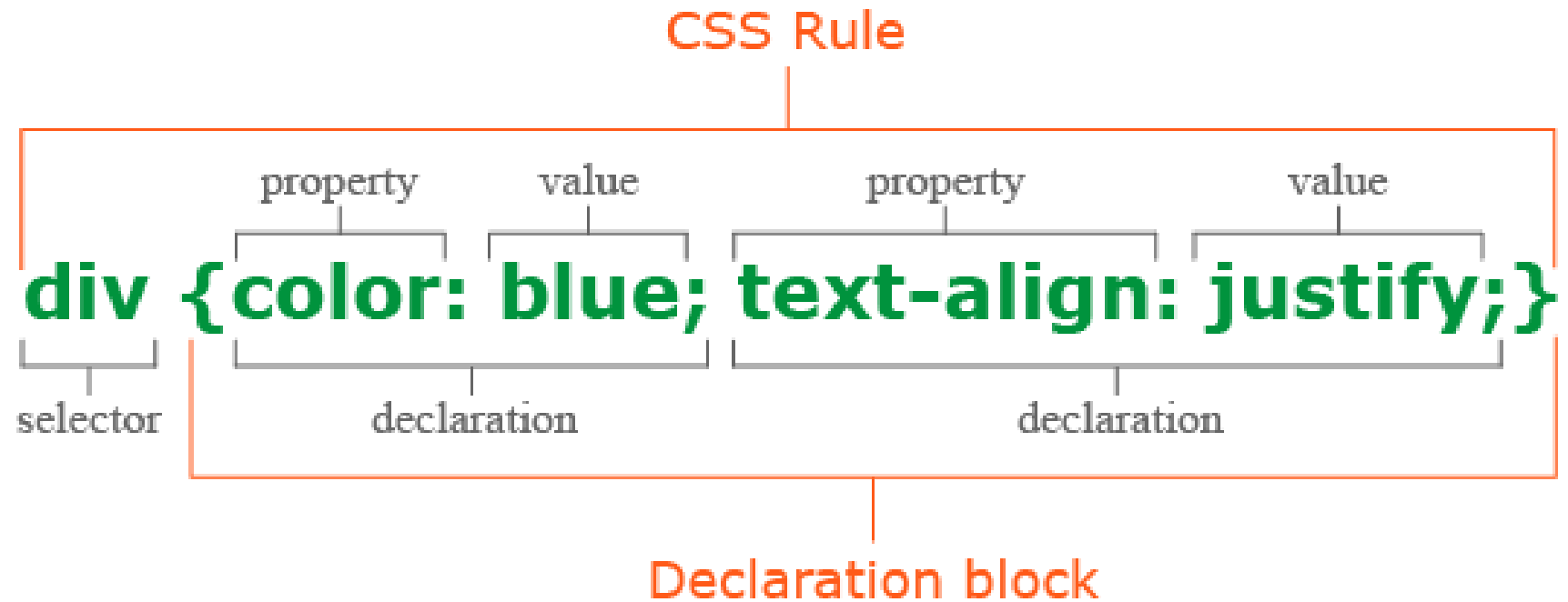
What is CSS?

- CSS stands for Cascading Style Sheets
- It's a style sheet language used for describing the presentation of a document written in HTML or XML.

Importance of CSS

- CSS plays a crucial role in separating content from presentation.
- It allows developers to style web pages, control layout, and enhance user experience.
- CSS saves a lot of work. It can control the layout of multiple web pages all at once

Basic Syntax



Internal, External, and Inline CSS

- Internal CSS: Styles are defined within the HTML document using the `<style>` tag in the `<head>` section.
- External CSS: Styles are defined in a separate CSS file and linked to the HTML document using the `<link>` tag.
- Inline CSS: Styles are applied directly to individual HTML elements using the `style` attribute.

```
<html>
<head>
  <!-- Internal CSS -->
  <style>
    p {
      color: blue;
      font-size: 16px;
    }
  </style>

  <!-- External CSS -->
  <link rel="stylesheet" href="styles.css">
</head>
<body>
  <!-- Inline CSS -->
  <p style="font-weight: bold;">This paragraph has inline CSS.</p>

  <p>This paragraph is styled using internal CSS.</p>
</body>
</html>
```

Selectors

- Element Selector: Targets HTML elements based on their tag name (e.g., `<p>` for paragraphs, `<h1>` for headings).
- Class Selector: Targets HTML elements with a specific class attribute using the `.class` notation (e.g., `.highlight`).
- ID Selector: Targets a single HTML element with a specific ID attribute using the `#id` notation (e.g., `#header`).

Selectors

```
/* Element Selector */  
p {  
    color: blue;  
}  
  
/* Class Selector */  
.highlight {  
    font-size: 20px;  
    background-color: yellow;  
}  
  
/* ID Selector */  
#header {  
    margin: 20px;  
    padding: 10px;  
}
```

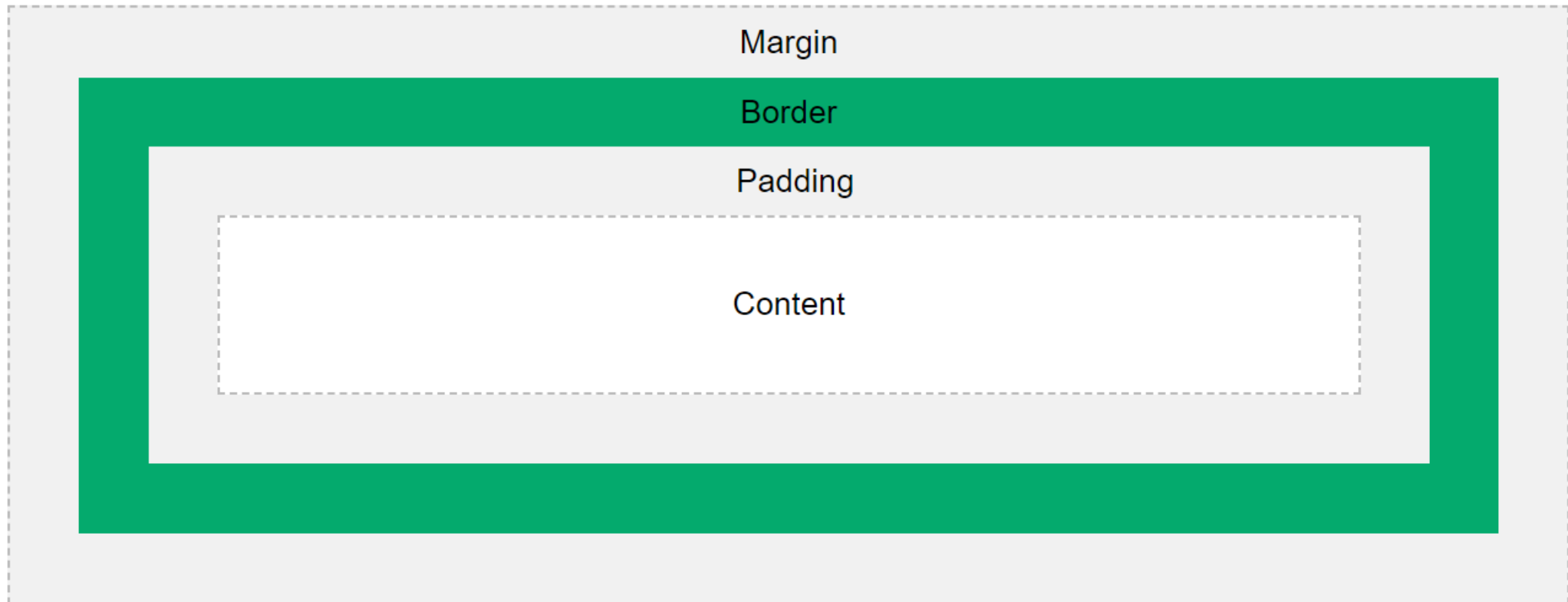
Example

```
p {  
  font-family: verdana;  
  color: red;  
  font-size: 20px;  
  text-align: center;  
}
```

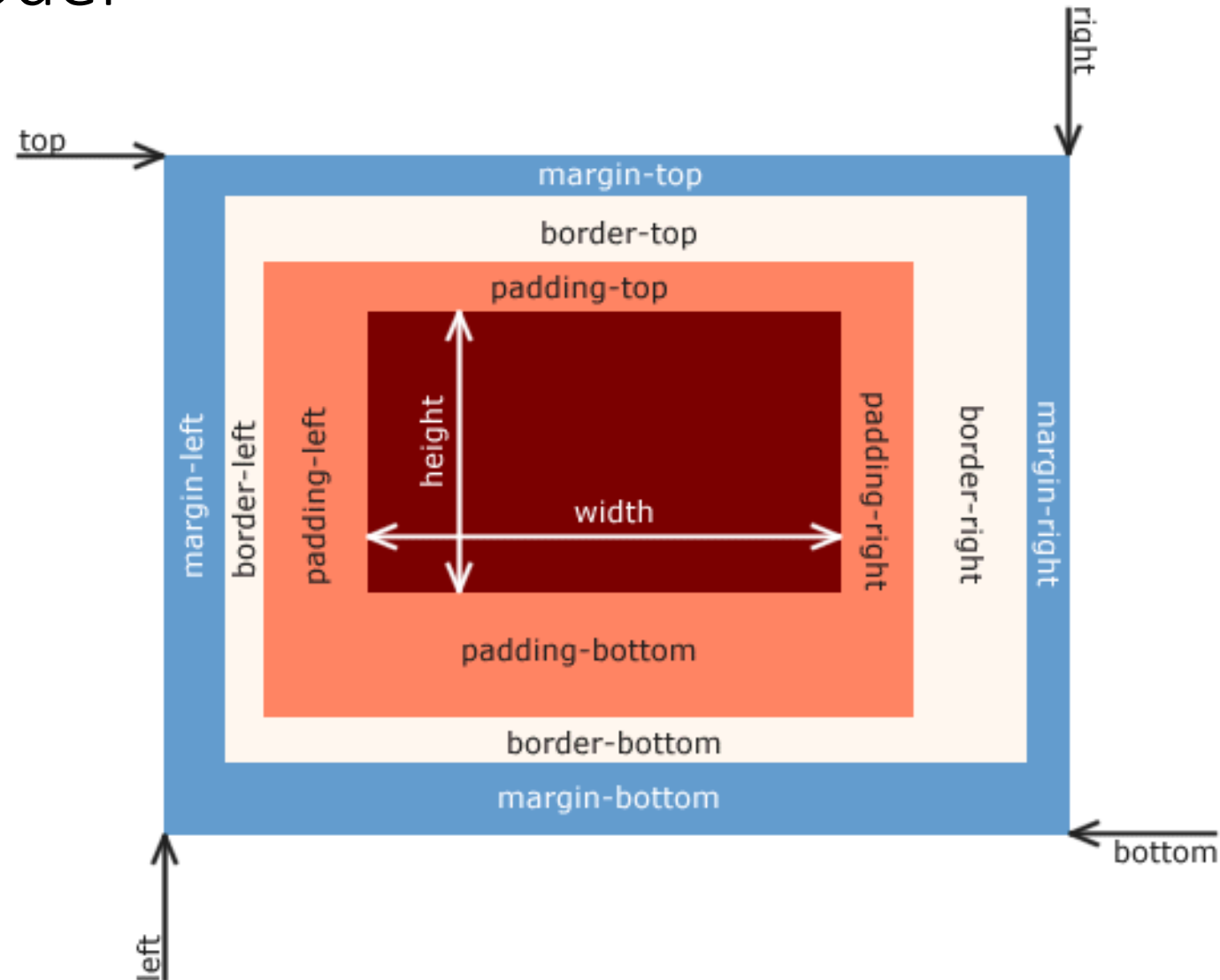
Common CSS Properties

- Color: Sets the color of text
- Font-Size: Specifies the size of the font
- Background-Color: Sets the background color of an element
- Margin: Controls the space outside of an element
- Padding: Controls the space inside of an element

CSS Box Model

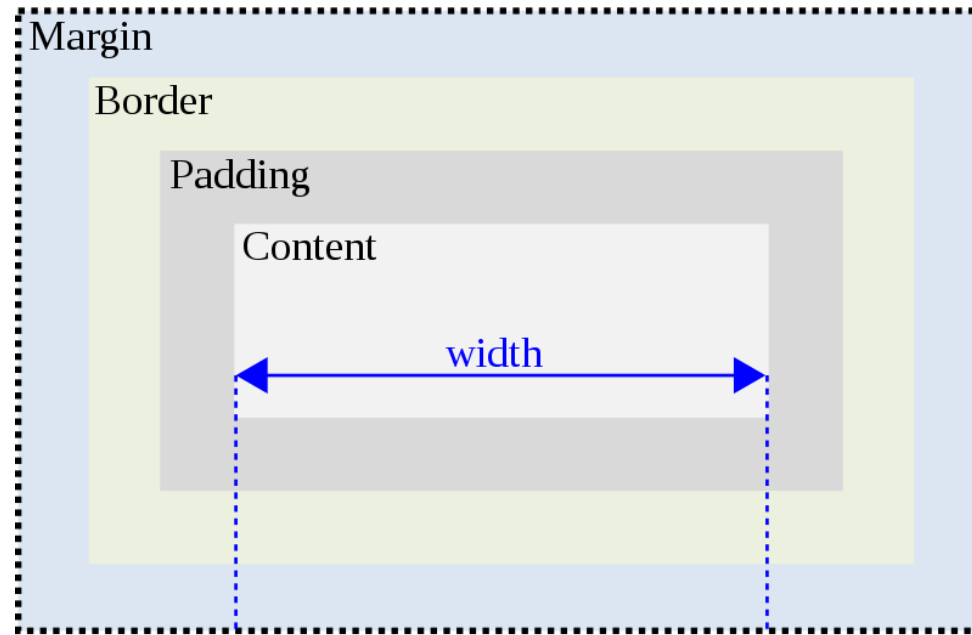


CSS Box Model

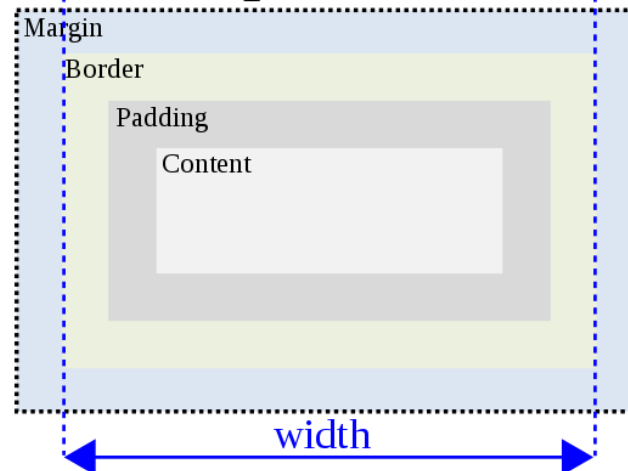


History

W3C box model



Internet Explorer box model



https://en.wikipedia.org/wiki/CSS_box_model

Example

```
div {  
    background-color: lightgrey;  
    width: 300px;  
    height: 180px;  
    border: 15px solid maroon;  
    padding: 50px;  
    margin: 20px;  
}
```

CSS Layout

- CSS provides various layout techniques to arrange elements on a web page effectively.
- Some common layout techniques include:
- Float: Elements are floated to the left or right within their container, allowing other elements to wrap around them.
- Position: Elements can be positioned relative to their containing element or to the viewport.
- Flexbox: A one-dimensional layout model that allows elements to be aligned and distributed within a container along a single axis.
- Grid: A two-dimensional layout model that enables precise placement of elements in rows and columns.

CSS Transition and Animation

- Transitions:
 - Allow smooth transitions between different property values over a specified duration.
 - They can be triggered by user interactions or state changes.
- Animations:
 - Provide more complex and customizable animations by defining keyframes and animation properties.
 - Animations can be used to create effects such as fading, sliding, or rotating elements.

Responsive Web Design

- Responsive design ensures that a website looks and functions well across various devices and screen sizes.
- Responsive design can result in improved visibility, higher click-through rates, and better search engine optimization (SEO)
- Reducing Bounce Rates: Responsive design helps minimize bounce rates (the percentage of visitors who leave a website after viewing only one page) by ensuring that users can easily access and navigate content on any device.

Responsive Web Design

- The viewport is the user's visible area of a web page.
- Setting the viewport:

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

CSS Frameworks

- Bootstrap: A popular front-end framework that provides a comprehensive set of pre-styled components, layout grids, and utilities for building responsive websites quickly.
- Foundation: Another robust front-end framework offering customizable components, responsive grid system, and flexible styling options.
- Benefits: CSS frameworks expedite development by providing pre-built components and styles, ensuring consistency, and simplifying responsive design implementation.

Best Practices

- Clean and maintainable CSS code enhances readability, makes it easier to understand and modify, and reduces the risk of introducing bugs.
- Test CSS layouts and features in multiple browsers and versions to identify and address any compatibility issues.